

Supervision-Matched Selection of Paired RAG Repairs: An Empirical Study of Accuracy and Damage

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Abstract

Retrieval-augmented generation (RAG) repair can recover an incorrect answer but can also replace an answer that was already correct. We study this trade-off after both candidate answers have been generated, holding candidate pairs, downstream supervision, eligibility, and replacement count fixed. The evaluation uses one Qwen2.5-3B-Instruct reader, 6,000 questions from three multi-hop question-answering datasets, three retrieval conditions, and nine selection policies. Each switching policy selects 900 of 18,000 traces. A calibrated logistic selector combines a frozen histogram-based gradient-boosting (HGB) score with a paired adaptation of Generate but Verify (GbV). Relative to a matched GbV-only selector, it improves exact match by 0.2333 percentage points and reduces full-population Damage by 0.0722 points; multiplicity-adjusted question-cluster bootstrap ranges are $[0.0722, 0.4333]$ and $[-0.1444, -0.0278]$. Against the matched HGB-only selector, observed Damage decreases from 26 to 11 events, but the joint accuracy-improvement and Damage-reduction rule is not met. A 25-feature selector also does not establish an advance over the two-signal selector. These comparisons show that a signal combination’s supported value depends on the matched control, and that similar final accuracies can conceal different recovery–damage profiles. The findings are conditional on the evaluated reader, candidate pipeline, and operating point; they do not establish a new architecture, cross-reader transfer, or deployment-cost savings.

Keywords retrieval-augmented generation; answer verification; post-generation selection; reliability; empirical evaluation

1 Introduction

Retrieval-augmented generation (RAG) supplies a language model with external evidence, but retrieval is not uniformly helpful [12, 13, 17]. A second retrieval pass can supply missing information and recover an incorrect answer. It can also introduce distracting evidence and replace an answer that was already correct [1]. Generating a repair candidate and deciding whether to adopt it are therefore different problems. The latter matters whenever a question-answering system already has an initial response and a revised alternative: the existence of new evidence does not itself justify replacing the response.

This paper studies the adoption decision after candidate acquisition. For each question and retrieval condition, a fixed pipeline generates an original answer a_0 and a repaired answer a_1 . A policy either keeps a_0 or switches to a_1 . The empirical question is whether combining a frozen internal score with an adapted external verifier improves correctness while reducing harmful replacements, relative to either signal alone, under shared candidates and an equal number of replacements.

Two distinctions motivate the comparison. First, final exact-match accuracy is the net outcome of useful recoveries and harmful replacements; it need not reveal how many previously correct answers were damaged. Second, a raw score and a fitted selector using that score differ in more than their inputs. A comparison that changes both the signal and the downstream training procedure cannot isolate the signal’s increment at that layer. We therefore compare five selection heads under the same fitting/calibration partitions, preprocessing, eligibility rule, and global action allocation.

Throughout this paper, *supervision-matched* means matched downstream supervision. The upstream scores have different historical training dependencies, which are not equalized. The comparisons are conditional on those frozen dependencies; they are not comparisons at an equal total historical labeling or training cost. HGB denotes an upstream histogram-based gradient-boosting score, whereas $\text{HGB}+\text{GbV}_R$ denotes a current logistic selection head receiving that score and the paired verifier margin.

Adaptive retrieval, verification-triggered editing, faithfulness prediction, and post-generation arbitration are established research directions [2, 4, 6, 9, 11, 19, 20, 22]. Paired evidence interventions have also been used to audit reader-conditional behavior [5]. Accordingly, we use ordinary calibrated logistic regression as a common empirical comparison instrument, not as a new selector architecture. Prior work already predicts

benefit between generated answers; the contribution here is a controlled assessment of scoring signals in the original-RAG versus repaired-RAG setting.

The study addresses three connected questions: whether the combination meets a joint accuracy-improvement and Damage-reduction criterion against each fitted single-signal control; whether expanding the feature set establishes an advance over the two-signal selector; and what the recovery–damage decomposition reveals beyond final accuracy. Five fitted heads and four contextual policies are evaluated on the same candidate pairs. Fixing both the downstream training recipe and 900-action allocation separates the evaluated signal increment from changes in that recipe or in intervention frequency.

The resulting evidence is comparison-dependent. Adding HGB to the matched GbV-only head passes the joint rule, whereas adding GbV to the matched HGB-only head does not. The largest feature set has the highest point accuracy but does not establish an advance under the same criterion. These findings, together with full-population Recovery and Damage accounting, are the empirical contribution. They do not make selection itself new, establish a formal difference between the two signal increments, or extend beyond the evaluated reader and candidate pipeline.

2 Related work

2.1 Adaptive and corrective RAG

Adaptive-RAG selects among no-, single-, and multi-step retrieval using question complexity [9]. Self-RAG trains a generator to retrieve and critique its outputs [2], while Corrective RAG evaluates retrieval quality and invokes corrective actions [20]. Verify-and-Edit checks reasoning chains, retrieves evidence, and edits low-consistency outputs [22]. D2R-RAG diagnoses failures and chooses repairs under resource constraints [7]; Doctor-RAG localizes and repairs failed agentic trajectories while reusing valid prefixes [10]. These systems change acquisition or generation. Our policies instead receive two already generated answers and select the final response. The 5% allocation limits replacements, not retrieval or generation compute.

2.2 Predicting benefit and post-generation arbitration

Self-Knowledge Guided Retrieval learns when external retrieval is helpful [17]. Dado et al. predict the benefit of RAG over a no-RAG answer using retrieval, answer, and semantic-consistency signals [4]. Their version 3 includes a paired natural-language-inference (NLI) difference, the two-answer Bert-Gen-2A predictor, and predicted-gain-driven Selective RAG evaluation. TrustMargin selects between Direct and RAG answers using likelihood views without training [19]. Counterfactual supervision has also been used to route search from paired no-search and forced-search outcomes [11].

Our candidate domain is original-RAG versus repaired-RAG. The distinction is not the availability of two answers or the use of their score difference, but the assessment made after fixing candidate pairs, downstream training, eligibility, and switch count. The fitted single-signal heads are the controls for this within-protocol attribution question. The cited studies are conceptual comparators: we report no numerical comparison with their complete systems and make no superiority claim over them.

2.3 Verification and retrieval harm

Generate but Verify introduces Answering with Faithfulness and evaluates faithfulness prediction for downstream use [6]. We adapt its entailment-based score to both completed candidate branches and use the branch-score difference as one selector input. Because the outcomes here are normalized answer correctness rather than faithfulness labels, this component is a paired GbV adaptation, not a direct replication of the original faithfulness evaluation. Studies of detrimental and distracting passages motivate explicit harm accounting [1, 13]; Pair-ID further emphasizes reader-conditional responses to evidence interventions [5]. These distinctions delimit the present inference: it concerns the fixed reader and realized original/repaired candidate pairs.

3 Experimental design and selection policies

3.1 Evaluation population and candidate generation

The evaluation contains 2,000 questions each from HotpotQA [21], 2WikiMultiHopQA [8], and MuSiQue [16]. Selection uses a prospectively fixed deterministic hash order after recorded-ID exclusion; it is not a probability sample. Each question is crossed with BM25 [15], BGE dense retrieval [18], and hybrid reciprocal-rank fusion (RRF) [3]. This yields 6,000 question groups and 18,000 traces, rather than 18,000 independent questions. The three retriever-specific traces for a question remain together in resampling. Table 1 summarizes the fixed design.

Table 1 Fixed study design. Document pools are dataset-specific.

Component	Setting
Question groups / traces	6,000 / 18,000
Datasets	HotpotQA, 2WikiMultiHopQA, MuSiQue; 2,000 questions each
Retrievers	BM25, BGE dense, hybrid RRF
Document pools	19,352; 11,746; 23,618 documents (54,716 total)
Reader	Qwen2.5-3B-Instruct
Downstream development	4,500 questions / 13,500 traces; 3,600 fitting and 900 disjoint calibration questions
Eligible evaluation traces	4,267 of 18,000
Action allocation	Global $K = 900$ for each switching policy; Keep takes zero actions
Primary resampling	20,000 dataset-stratified question-cluster draws

Each dataset has a deduplicated corpus constructed from selected benchmark contexts without reading answers, supporting facts, or decompositions. The pools contain 19,352 HotpotQA, 11,746 2WikiMultiHopQA, and 23,618 MuSiQue documents. They are bounded benchmark-derived pools, not full Wikipedia. The public source splits and historical/current-development separation are stated in Appendix G.

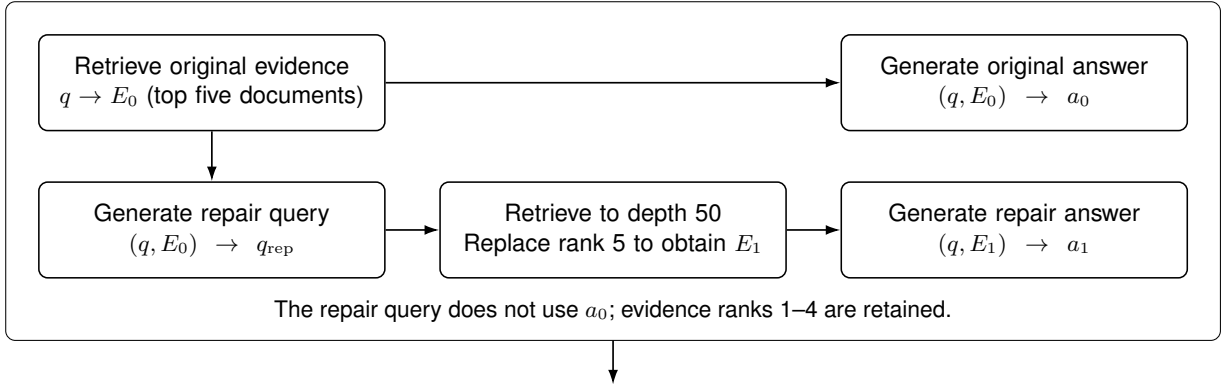
For each trace, the pipeline retrieves the top five documents E_0 and generates a_0 from the question and E_0 . It then generates a repair query from the question and E_0 , without using a_0 , and retrieves to depth 50 with the same retriever. The first document absent from all five documents in E_0 replaces the document at rank 5; ranks 1–4 are retained. The reader generates a_1 from the resulting evidence E_1 . Every policy receives this same (a_0, a_1) pair. Figure 1 separates shared acquisition from subsequent selection.

The reader is Qwen/Qwen2.5-3B-Instruct [14], pinned at revision aa8e72537993ba99e69dfaafa59ed015b17504d1. The dense encoder is BAAI/bge-base-en-v1.5 at revision a5beb1e3e68b9ab74eb54cfd186867f64f240e1a. Generation uses the frozen chat template and task prompts, greedy decoding, seed 20260828, at most 48 answer tokens, at most 64 repair-query tokens, and a 16,000-character evidence budget. The study record reports an exact acquisition replay on 180 prospectively selected traces, rather than a second complete acquisition. The original task-prompt texts are absent from the released materials, which limits reconstruction of candidate acquisition (Appendix J).

3.2 Scoring signals and selection policies

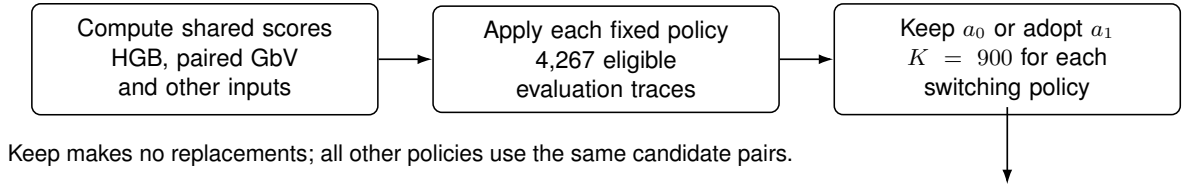
All policy inputs are calculated without access to the fresh evaluation reference answers. This does not imply that the upstream estimators were trained without labels. The shared score matrix contains 11 numeric inputs: seven outputs from frozen historical estimators (state-symmetric HGB, state-symmetric logistic, no-cross-state, no-B, no-evidence-change, no-answer-form, and ordinary compact logistic); three fixed rules (B-rule, higher-own-likelihood, and likelihood margin); and the paired GbV margin. Historical identifiers are retained to map the reported panel to its saved artifacts; their names alone do not establish a component’s necessity.

01 Shared candidate generation



02 Fixed-batch answer selection

Fitted heads share 3,600 fitting and 900 disjoint calibration questions.



Evaluation references are accessed only after action memberships are fixed.

03 Evaluate all 18,000 traces

EM and token F1; Recovery, Damage and Net
20,000 question-cluster bootstrap draws; global top- K is reallocated within each draw

Fig. 1 Candidate generation, answer selection and evaluation. All policies receive the same original and repaired answers. HGB denotes histogram-based gradient boosting and GbV denotes the paired Generate but Verify adaptation. The repair query does not use the original answer. Evaluation references are accessed only after all action memberships are fixed; prior upstream training is not label-free.

HGB-based preference score. HGB is a historical pairwise selector, not a model fitted on the current 4,500-question development cohort. Let L_{ij} denote the reader’s mean teacher-forced log probability for answer a_i under evidence E_j . HGB receives 48 features comprising eight repaired-minus-original branch differences and their interactions with five pair-context variables. Its historical target is one for $(EM(a_0), EM(a_1)) = (0, 1)$ and zero for $(1, 0)$; same-correctness transitions are excluded from that historical fit. Each (x, y) is augmented with $(-x, 1 - y)$ and the resulting probabilities are symmetrized at scoring time. A larger score supports choosing a_1 . Appendix E gives the feature construction, training parameters, score, and provenance boundary. Its symmetry is a property under feature-vector negation; it is not asserted to be equivariance under exchanging the raw answer–evidence branches and recomputing all features.

Paired GbV margin. For branch i , each evidence passage is a premise and the question and answer form the hypothesis

The answer to the question "{question}" is: "{answer}"

The frozen implementation uses this template for both branches, including the question rather than testing an isolated answer string. An overlength premise is split into word windows, with 20-word overlap subject to forward progress; each complete premise–hypothesis pair, including special tokens, is checked against the 512-token limit. The hypothesis is retained. No silent pair truncation is performed: an unfit hypothesis or

even a single unfit premise word raises an error in this scorer. Appendix D specifies the documented input and chunking interface.

For each branch, let $F_i(q, a_i, E_i)$ be the maximum entailment probability over its evidence chunks. The input is

$$g = F_1(q, a_1, E_1) - F_0(q, a_0, E_0). \quad (1)$$

The verifier is `MoritzLaurer/deberta-v3-large-zeroshot-v2.0` at revision `5a4338ab2151dc8db04ad53b42b6153382bf4f99`, evaluated in FP32, with entailment label index 0. The branch difference, pinned runtime, common eligibility, and global allocation are project choices, not an author-supplied paired-policy implementation. Base scoring also uses BGE answer-pair embeddings and four Qwen answer-token-only likelihood cells with an 8,192-token scoring limit.

Policy panel. The nine policies are Keep, raw HGB, raw paired GbV, ROA-FULL, ROA-NOGBV, HGB+GbV_R, HGB-only_R, GbV-only_R, and a sealed historical V2 policy. The suffix *R* denotes a fitted Recovery head. ROA-FULL has 11 numeric inputs, their 11 missingness flags, and three retriever indicators (25 columns). ROA-NOGBV removes the GbV input and its flag (23 columns). HGB+GbV_R uses HGB and the GbV margin, their flags, and the same retriever indicators (7 columns). The matched single-signal heads each use one score, one flag, and three indicators (5 columns). Thus, “two-signal” means two numeric score sources, not a two-column design matrix. The construction map is given in Appendix D.

Raw HGB directly ranks by the historical score. HGB-only_R fits a current logistic Recovery head to that score, while HGB+GbV_R adds the verifier margin. Neither refits the historical HGB estimator. Raw HGB, raw GbV, and V2 have different upstream training or rule dependencies and are contextual comparisons, not supervision-matched attribution controls. Keep is the no-action reference.

3.3 Downstream training and action allocation

The five fitted heads share 4,500 development questions and 13,500 traces. A deterministic within-dataset hash order assigns 3,600 questions (10,800 traces) to fitting and 900 questions (2,700 traces) to disjoint calibration, with retriever siblings kept together. Eligibility is then applied inside each partition. Every base head therefore fits 2,572 eligible traces: 2,015 negatives and 557 Recovery positives. Every Platt calibrator uses 630 eligible traces: 498 negatives and 132 positives. These are the effective numerical sample sizes; 10,800 and 2,700 are partition population counts, not the numbers of rows entering the fits.

For an eligible development trace, the target is

$$y_R = \mathbf{1}\{\text{EM}(a_0) = 0, \text{EM}(a_1) = 1\}. \quad (2)$$

Other transitions are negative for this target. Unlike the historical HGB preference target, this definition does not discard neutral transitions that otherwise satisfy eligibility. Numeric missing values are imputed using eligible fitting-set medians. Means and population standard deviations are then computed on the imputed fitting matrix; zero standard deviations are replaced by one. The same transformation is used for calibration and evaluation. Missingness flags and BM25/dense/hybrid indicators are unscaled. Dataset ID, sample ID, answer text, reference strings, and future actions are not additional direct predictor columns.

Each base head is L2-regularized logistic regression with $C = 1$, `lbfgs`, no class weights, `max_iter=5000`, and tolerance 10^{-4} . A separate Platt logistic model uses the disjoint calibration-set logits, $C = 10^6$, `lbfgs`, `max_iter=2000`, and tolerance 10^{-4} . The five base/calibration bundles require ten fit calls. Both models include an intercept; implementation identifiers and saved parameter records are specified in Appendix D.3. No head is refitted after the fresh evaluation outcomes are opened.

Writing a standardized design row as v and the base logit as $\ell = w^\top v + b$, the fixed-batch ranking score has the form

$$s(v) = \sigma(\alpha\ell + \beta), \quad \sigma(t) = (1 + e^{-t})^{-1}. \quad (3)$$

Here α and β are estimated only from the calibration partition. A positive, strictly monotone calibration preserves logit order in exact arithmetic, although finite-precision ties can affect membership. We therefore do

not attribute the observed replacement gains to calibration itself. The recorded policy always ranks by its calibrated score and uses the stated tie rule.

All 18,000 evaluation traces remain in the denominator. The common eligibility mask contains 4,267 traces. The 13,732 traces with equal normalized answers and one trace with an empty original answer are forced to Keep for every policy. Each switching policy ranks eligible traces by its frozen score in descending order and selects

$$K = \text{round}(0.05 \times 18,000) = 900. \quad (4)$$

Ties are resolved by (dataset, retriever, sample_id). There are no dataset/retriever quotas, tuned thresholds, or post-outcome action changes. This is a fixed experimental operating point for comparing replacement behavior at equal action count; it is not claimed to be the optimal budget. Five percent refers to the complete population, not to the eligible subset or the computational workload.

Recovery is an empirical ranking target, not a guarantee of optimal net benefit. For one proposed replacement, let $c_i = \text{EM}(a_i) \in \{0, 1\}$. The expected binary correctness gain is

$$\mathbb{E}[c_1 - c_0 \mid v] = P(\text{Recovery} \mid v) - P(\text{Damage} \mid v). \quad (5)$$

This identity explains why ranking by Recovery probability alone need not rank by expected net gain. The selector uses the Recovery target rather than a direct-net objective or an explicit Damage constraint. Reduced Damage is therefore an evaluated outcome, not a training guarantee.

3.4 Outcome evaluation and statistical analysis

Candidate acquisition, all scores, common eligibility, and complete action memberships were sealed before evaluation-reference access. A separate mapper evaluated 6,000 selected reference sets (6,863 strings) against 36,000 fixed answers and emitted numeric EM/F1 outcomes. An independent method-blind implementation recomputed the 72,000 scalars in the study record. This establishes the documented order within the recorded execution graph, not the absence of unknown activity outside that graph.

For a policy’s selected replacements, Recovery counts ($\text{EM}(a_0), \text{EM}(a_1) = (0, 1)$), Damage counts (1, 0), and Neutral contains the two remaining transitions. Neutral need not imply unchanged token F1. With $N = 18,000$,

$$\text{Net} = \text{Recovery} - \text{Damage}, \quad (6)$$

$$\Delta \text{EM}_{\text{Keep}} = 100 \text{ Net} / N \quad \text{percentage points}, \quad (7)$$

$$\text{Damage rate} = 100 \text{ Damage} / N \quad \text{percent}. \quad (8)$$

EM, F1, and Damage rate use the complete population. Damage is not normalized by the number of actions or originally correct answers in the reported primary endpoint. Counts and rates are reported together; normalization and multiple-reference handling appear in Appendix F.

The frozen primary family comprises three ordered comparisons with two endpoints each: EM difference and Damage-rate difference. We use 20,000 dataset-stratified question-cluster bootstrap draws with NumPy `default_rng(20260926)`. All three retriever-specific traces remain with each sampled question, and global top- K allocation is repeated within each draw. Percentile endpoints use quantiles 0.05/12 and $1 - 0.05/12$, with linear interpolation, applying a two-sided Bonferroni adjustment to the six-endpoint family. A comparison passes the joint rule only if the adjusted EM range is strictly above zero and the adjusted Damage range is strictly below zero.

The pre-specified fixed-action sensitivity reuses the same draws but retains observed memberships; it does not require exactly 900 action copies in each resampled population (Appendix A). Both analyses condition on the fitted heads and realized candidate pools. They exclude uncertainty from fitting, calibration, generation, and pool construction, and do not provide exact finite-sample coverage or guarantees for new readers or domains.

4 Results

4.1 Results for all selection policies

Table 2 reports all nine policies. Keep obtains 17.9889% EM and 22.7207% token F1. Each switching policy performs 900 replacements. Their observed gains over Keep range from 1.6278 to 1.9389 EM percentage points, while full-population Damage rates range from 0.0611% to 0.2111%. These Keep comparisons are descriptive and are outside the frozen primary interval family.

HGB+GbV_R records 352 Recoveries and 11 Damage events, giving Net = 341, EM = 19.8833%, and F1 = 25.2780%. It has the lowest observed Damage among the eight equal-action switching policies. Keep has zero Damage because it makes no replacements and is not part of that equal-action ranking. ROA-FULL has the highest point EM and F1, but its 371 Recoveries accompany 22 Damage events. The two panels of Table 2 distinguish event counts from answer-quality rates for the same policies and denominator. Figure 2 makes the corresponding recovery–damage profiles visible.

Table 2 Complete fixed-allocation results. Panel A contains trace-event counts. Panel B gives percentages and differences from Keep in percentage points (pp). All rates use $N = 18,000$; each switching policy makes 900 replacements.

Panel A. Replacement events					
Policy	Actions	Recovery	Damage	Neutral	Net
Keep	0	0	0	0	+0
HGB	900	366	32	502	+334
GbV	900	318	25	557	+293
ROA-FULL	900	371	22	507	+349
ROA-NOGBV	900	355	38	507	+317
HGB+GbV _R	900	352	11	537	+341
HGB-only _R	900	357	26	517	+331
GbV-only _R	900	323	24	553	+299
V2	900	361	28	511	+333

Panel B. Answer quality and full-population Damage					
Policy	EM (%)	Δ EM (pp)	F1 (%)	Δ F1 (pp)	Damage (%)
Keep	17.9889	+0.0000	22.7207	+0.0000	0.0000
HGB	19.8444	+1.8556	25.1593	+2.4386	0.1778
GbV	19.6167	+1.6278	24.8928	+2.1722	0.1389
ROA-FULL	19.9278	+1.9389	25.2988	+2.5781	0.1222
ROA-NOGBV	19.7500	+1.7611	25.0321	+2.3115	0.2111
HGB+GbV _R	19.8833	+1.8944	25.2780	+2.5574	0.0611
HGB-only _R	19.8278	+1.8389	25.1323	+2.4117	0.1444
GbV-only _R	19.6500	+1.6611	24.9344	+2.2137	0.1333
V2	19.8389	+1.8500	25.1712	+2.4506	0.1556

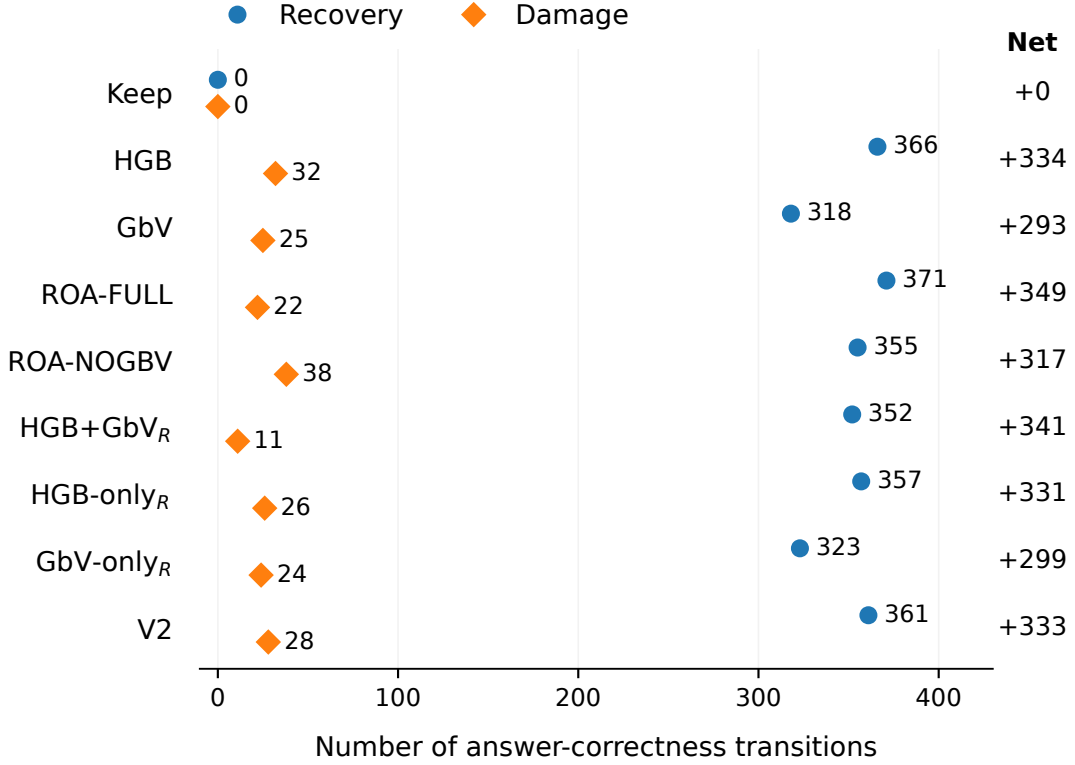


Fig. 2 Recovery and Damage counts for all nine policies, with net gains in the right-hand column. Circles denote Recovery and diamonds denote Damage. All switching policies perform 900 replacements; Keep performs none. Counts refer to the complete 18,000-trace population and reproduce Table 2. No uncertainty intervals are implied.

4.2 Primary comparisons

Table 3 reports every pre-specified ordered comparison and both endpoints; Figure 3 visualizes their adjusted ranges. Relative to GbV-only_R, HGB+GbV_R improves EM by 0.2333 percentage points and reduces Damage by 0.0722 points. The adjusted ranges, [0.0722, 0.4333] and [−0.1444, −0.0278], exclude zero in the required directions. This comparison passes the joint rule and corresponds to 42 more correct final-answer traces and 13 fewer Damage events across 18,000 traces. It supports an increment from the frozen HGB signal when added to the matched GbV-only head under this protocol.

Table 3 Primary ordered comparisons. The two-sided multiplicity-adjusted percentile ranges use 20,000 dataset-stratified question-cluster draws, with global top-*K* reallocated in each draw. All numerical values are percentage points.

Ordered comparison	Endpoint	Difference	Adjusted range	Joint rule
ROA-FULL	Δ EM	+0.0444	[−0.1333, +0.2037]	Not met
minus HGB+GbV _R	Δ Damage	+0.0611	[+0.0000, +0.1278]	
HGB+GbV _R	Δ EM	+0.0556	[−0.1556, +0.3167]	Not met
minus HGB-only _R	Δ Damage	−0.0833	[−0.1833, −0.0056]	
HGB+GbV _R	Δ EM	+0.2333	[+0.0722, +0.4333]	Met
minus GbV-only _R	Δ Damage	−0.0722	[−0.1444, −0.0278]	

Relative to HGB-only_R, the combination changes EM by 0.0556 points. The adjusted EM range is [−0.1556, 0.3167]. Its Damage difference is −0.0833 points, with range [−0.1833, −0.0056]. The Damage endpoint favors the combination in the primary analysis, but the EM interval crosses zero; the joint rule does not pass. The underlying counts are 352 rather than 357 Recoveries and 11 rather than 26 Damage events: five fewer recoveries and fifteen fewer harmful replacements give ten additional correct final-answer traces.

This is evidence of a different observed replacement profile, not evidence of joint improvement or accuracy non-inferiority.

For ROA-FULL minus HGB+GbV_R, the EM difference is 0.0444 points $[-0.1333, 0.2037]$ and the Damage difference is +0.0611 points $[0.0000, 0.1278]$. The joint rule is not met and the observed Damage direction is unfavorable. Its nineteen extra Recoveries accompany eleven extra Damage events, giving eight additional correct final-answer traces relative to the two-signal selector. A higher point EM therefore does not establish an advance under the declared two-endpoint criterion.

The single-signal comparisons examine different increments. One adds HGB to a GbV-only head; the other adds GbV to an HGB-only head. One passing result and one non-passing result are neither an interaction test nor proof that the increments differ causally. The non-passing HGB-only comparison also does not show that GbV is unnecessary: no equivalence or non-inferiority design was used.

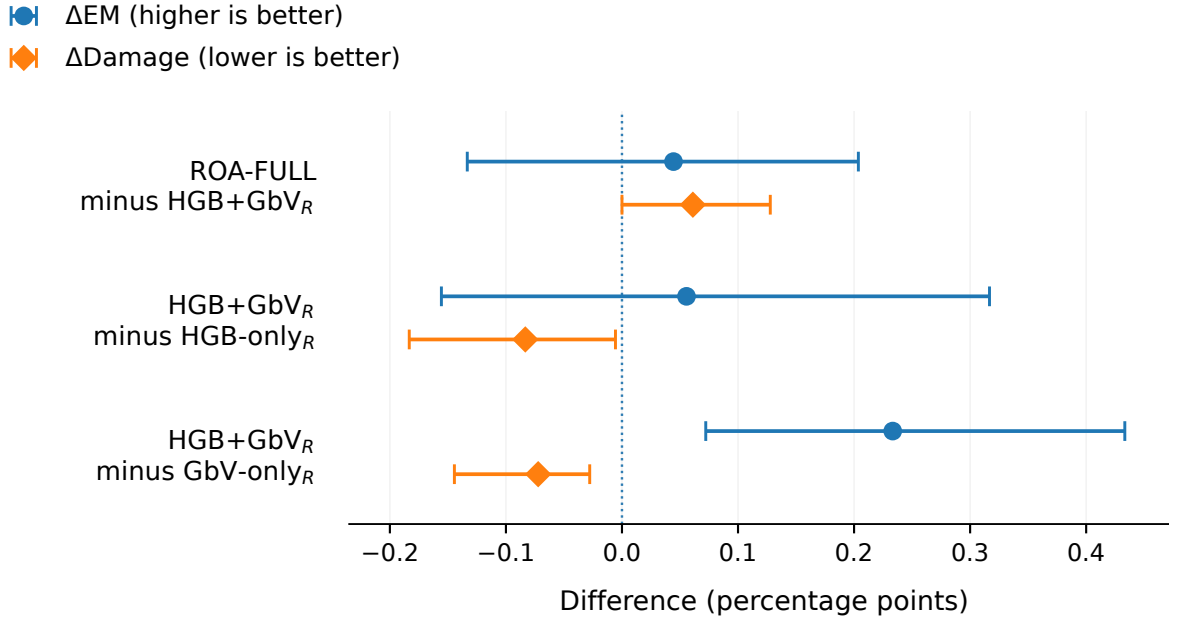


Fig. 3 Primary point differences and multiplicity-adjusted percentile ranges reported in Table 3. Circles denote EM differences and diamonds denote full-population Damage-rate differences. Positive EM and negative Damage differences favor the first policy. The dotted line marks zero; only the comparison against GbV-only_R meets both parts of the joint criterion.

4.3 Sensitivity analysis and subgroup results

Holding observed action memberships fixed preserves the HGB+GbV_R versus GbV-only_R joint decision: adjusted EM and Damage ranges are $[0.0389, 0.4278]$ and $[-0.1389, -0.0222]$. The HGB-only comparison remains non-passing; its EM range crosses zero and the Damage upper endpoint touches zero. The ROA-FULL comparison also remains non-passing. The complete sensitivity table appears in Appendix A.

Appendices B and C partition the same global allocation, rather than conducting independent 900-action experiments within each group. The combination’s Damage counts are 4, 7, and 0 across 2WikiMultiHopQA, HotpotQA, and MuSiQue, compared with 9, 12, and 5 for HGB-only_R. These are descriptive counts, not three independent confirmations of a harm-reduction claim.

The retriever breakdown also shows that score changes alter where actions are allocated. BM25 receives 300 combination actions versus 212 HGB-only actions, while dense retrieval receives 312 versus 395. In dense retrieval, the corresponding Net counts are 122 and 137. The aggregate improvement can consequently reflect both within-retriever ranking and between-retriever reallocation. These tables do not identify the relative contribution of those mechanisms, and no favorable subgroup is promoted to a superiority claim. Figures 2 and 3 visualize the full-population counts and existing primary ranges; neither introduces an additional analysis.

5 Computational workload and reproducibility

All policies consume a shared, precomputed acquisition and scoring graph. The candidate stage performs 18,000 original and 18,000 repair retrievals and records 54,000 Qwen generations. Both answers already exist before a policy makes its decision. Retaining replacements on 5% of traces therefore does not mean that only 5% of the acquisition work was performed.

Appendix I preserves all reported operation counts, token totals, stage timers, and separate memory peaks. The timers have different boundaries and are not added to form an end-to-end latency. Standalone policy deployment, per-trace latency distributions, a uniform end-to-end timer, energy, and floating-point operation counts were not measured. No deployment-efficiency or compute-saving comparison follows from the fixed action allocation.

The public materials support two complementary levels of reproduction from realized numerical inputs. The text-free development bundle contains 13,500 traces with opaque question-group identities, frozen fitting/calibration roles, eligibility, 11 score inputs, and binary EM outcomes. The current-head workflow refits five base models and five disjoint Platt calibrators and checks their outputs against the accepted preprocessing values, coefficients, iteration counts, target counts, and design hashes. The effective fitting and calibration sets contain 2,572 and 630 eligible traces, respectively.

The separate 18,000-trace evaluation bundle retains grouping, tie order, scores, actions, and numeric outcomes. Its workflow reconstructs all nine-policy point estimates and the complete fixed-seed, 20,000-draw question-cluster bootstrap, including reallocated top- K and fixed-action sensitivity. The release specifies exact equality with its saved numerical records. These workflows reproduce the current heads and statistical analysis from released numeric inputs; they do not regenerate retrieval, candidate answers, or neural scores. Appendix D.3 identifies the code and parameter records, and Appendix J states the full release boundary.

6 Limitations and threats to validity

6.1 Reader, candidate, and population scope

The reported effects concern one 3B-parameter Qwen reader, three multi-hop QA datasets, three fixed retrieval conditions, and bounded benchmark-derived document pools. They establish no transfer to other model families or sizes, unseen domains, full-corpus retrieval, or production workloads. The Phi extension failed its fixed semantic validation gate before an effect analysis; Mistral stopped before engineering and execution. Neither supplies cross-reader effect evidence (Appendix H).

The repair operator replaces only rank 5 after a generated repair query. The estimand conditions on both candidate answers already existing, the common eligibility rule, and a single global 900-action allocation. Different repair operators can change candidate quality and policy ranking. The study does not evaluate an online threshold policy, an adaptive action budget, or the optimality of the 5% operating point. The 18,000 traces increase the information available within this protocol; they are not independent evidence across different readers or repair mechanisms.

6.2 Metrics, inference, and mechanism

Normalized EM is strict and can miss semantically correct answer variants; token F1 is reported descriptively. Recovery and Damage are correctness transitions, not evidence-faithfulness labels. The GbV component remains an adaptation of a faithfulness verifier. Since the primary Damage rate uses all 18,000 traces, it must be read with the event counts and action count rather than interpreted as a conditional probability of harming an initially correct answer.

Question-cluster resampling retains within-question retriever dependence and stratifies datasets, but conditions on the fitted heads and realized pools. It does not include variation from refitting or regenerating candidates. Dataset/retriever cells are views of one global allocation, not independent replications. The results do not establish equivalence, non-inferiority, a universal fusion rule, or a causal explanation of signal complementarity. In particular, the positive HGB addition to a GbV-only head does not equalize the signals'

historical supervision costs. The HGB-only comparison’s favorable Damage endpoint in the primary analysis also does not justify claiming preserved accuracy; its EM interval includes zero and the fixed-action Damage interval touches zero.

6.3 Historical supervision and implementation provenance

The selected cohort has zero recorded dataset/ID overlap with the known 18,600-question exclusion union. ID separation is not semantic decontamination and cannot rule out pretraining exposure or unknown unlogged historical use. Benchmark-supplied contexts can also retain dataset-specific structure despite field-restricted corpus construction.

The seven historical estimators are frozen; their saved computations and reconstructed membership are documented. A locally preserved pre-ROA-named package contains a byte-identical historical training manifest, but its timestamp has no independent external certification. Original per-estimator fit-time ID/matrix receipts and an independent original-fit witness remain unavailable. Appendix E reports the reconstructed HGB fit and its numerical replay; this is not authentication of every original historical execution event.

Incomplete generation-prompt and legacy-component specifications, together with the absence of a complete upstream code-to-artifact release, limit end-to-end reconstruction (Appendix J). This limitation is distinct from the released current-head refitting and statistical reconstruction. Successful reconstruction from numeric inputs does not independently validate every upstream process that generated those inputs.

7 Discussion

7.1 Answer accuracy and harmful replacements

The event decomposition explains why similar final accuracies need not imply similar adoption behavior. Compared with HGB-only_R , HGB+GbV_R produces five fewer Recoveries and fifteen fewer Damage events, for ten additional correct final-answer traces. ROA-FULL produces nineteen more Recoveries than the two-signal selector, but eleven more Damage events, for eight additional correct final-answer traces. The small net differences conceal different compositions of beneficial and harmful actions. Equal action count prevents the number of interventions alone from explaining those compositions.

Within this panel, selecting by the highest point EM would identify ROA-FULL, whereas selecting by the lowest observed Damage among switching policies would identify HGB+GbV_R . Neither descriptive ranking settles the declared joint comparison: the combination passes that rule against GbV-only_R , but not against HGB-only_R , and ROA-FULL does not pass it against the combination. This is the practical information added by reporting event counts alongside final answer quality, rather than presenting a single “best method” ranking.

The implication is to make both correction and harmful replacement visible whenever already generated repair candidates are selectively adopted. It is not a claim that one observed trade-off is universally preferable. A system designer may value a harmful replacement differently from a missed recovery, and the present joint rule is not a fitted utility function or a safety guarantee. In particular, fewer observed Damage events cannot substitute for a demonstration of accuracy non-inferiority.

7.2 Incremental signal value under matched supervision

The supported positive result is the increment obtained by adding the frozen HGB signal to a matched GbV-only head. The reverse starting point does not establish joint improvement, even though the primary Damage endpoint favors the combination. These statements can coexist because they assess the combination against different controls. They should not be summarized as an unconditional benefit of signal fusion or as proof that the two incremental effects differ.

A comparison only with raw GbV would combine changes in score inputs with changes in the downstream fitting and calibration recipe. The fitted single-signal controls hold that recipe fixed, giving a narrower interpretation conditional on the same upstream dependencies. The scientific value of the controlled panel is therefore not simply a higher reported accuracy; it is the separation between what the combination establishes against one starting point and what remains unestablished against the other.

Conversely, the present matching does not equalize all historical labels, training effort, or acquisition costs. Nor does using a common calibrated logistic family demonstrate that calibration caused the gains. The evidence supports within-protocol incremental assessment, not a general claim that the combination is more sample-efficient, cheaper, or superior to the complete systems discussed in related work.

7.3 Feature-set expansion and subgroup interpretation

ROA-FULL has the highest point accuracy but does not establish an advance over the two-signal selector under the declared joint rule. This limits the claim for the particular evaluated feature stack; it does not show that additional features or complex selectors are generally unhelpful. The experiment does not distinguish redundancy, score quality, finite event counts, calibration, or candidate composition as explanations.

Similarly, the retriever breakdown makes action reallocation visible but does not isolate a causal mechanism. Because each head can use retriever indicators and receives one pooled action budget, the final counts can change through both within-group ranking and cross-group allocation. Describing that distinction is more defensible than attributing every avoided Damage event to a particular semantic verification ability. The dataset partitions likewise describe one shared allocation rather than independent confirmations.

For an intelligent question-answering system, the study provides a bounded evaluation pattern: fix the candidate pairs and downstream comparison conditions, count harmful replacements explicitly, and interpret each gain against the corresponding matched control. The released numerical bundles make the current-head fits and statistical comparisons inspectable within that boundary. Decisions about candidate acquisition, online operating points, and distribution shift remain outside the experiment and require evidence for the system’s own reader, corpus, and workload.

8 Conclusion

We examined post-generation selection between original and repaired RAG answers on 6,000 questions crossed with three retrievers, using a fixed Qwen2.5-3B-Instruct reader and shared candidate pairs. Under matched downstream supervision and a global 900-action allocation, HGB+GbV_R improves EM and reduces full-population Damage relative to GbV-only_R according to the pre-specified joint rule. The same rule does not pass against HGB-only_R, and ROA-FULL does not establish an advance over the two-signal selector.

The empirical contribution is a comparison-dependent assessment of scoring signals and an explicit account of recoveries and harmful replacements. Similar point accuracies can conceal different adoption behavior; neither the largest feature stack nor the highest point EM alone settles the declared comparison. Public development and evaluation bundles support reconstruction of the current heads and reported statistics from realized numerical inputs.

These conclusions remain conditional on the evaluated reader, candidate pipeline, and operating point. They do not establish a new selector architecture, accuracy non-inferiority, cross-reader transfer, or compute savings. The result is a controlled account of what the evaluated signals do and do not establish, rather than a universal prescription for RAG repair selection.

Data and code availability

Project-authored method and reporting code is publicly available at <https://github.com/qianxingji/ReliableRAG-Code> under Apache License 2.0. This manuscript is bound to commit 038d769e96d092a2eec3bbc5df9f45f3c2a17ff0. The release contains the historical HGB feature source, paired NLI scorer, current-head fitting code, accepted parameter records, a text-free 13,500-trace development bundle, and a text-free 18,000-trace evaluation bundle. The public workflow refits the five current heads, rebuilds all nine-policy point estimates and the complete 20,000-draw question-cluster bootstrap, and runs reporting and repository-integrity checks against the saved records.

The development fit has 2,572 eligible traces (557 positive) and calibration has 630 (132 positive). Benchmark text and original IDs, references, generated answers, retrieved passages, model weights, historical learned estimators, and private execution records are not redistributed. The release therefore does not

regenerate retrieval, candidate generation, neural scoring, or historical upstream training. Third-party data and models remain under their own terms. Appendices D.3 and J give the identifiers and the same reproducibility boundary.

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Appendix A Sensitivity to fixed action membership

The primary analysis reallocates global top- K actions within each question-cluster bootstrap draw. The pre-specified sensitivity analysis holds the observed action memberships fixed while reusing the same 20,000 draws and six-endpoint adjustment. It therefore does not force 900 action copies in each resampled population. Table 4 reports this fixed-membership sensitivity separately from the primary reallocation analysis.

Table 4 Fixed-action sensitivity. Observed action memberships are held fixed within the same 20,000 question-cluster draws and six-endpoint adjustment. All numerical values are percentage points.

Ordered comparison	Endpoint	Difference	Adjusted range	Joint rule
ROA-FULL minus HGB+GbV _R	Δ EM Δ Damage	+0.0444 +0.0611	[−0.1333, +0.2278] [+0.0000, +0.1278]	Not met
HGB+GbV _R minus HGB-only _R	Δ EM Δ Damage	+0.0556 −0.0833	[−0.2000, +0.3111] [−0.1722, +0.0000]	Not met
HGB+GbV _R minus GbV-only _R	Δ EM Δ Damage	+0.2333 −0.0722	[+0.0389, +0.4278] [−0.1389, −0.0222]	Met

Appendix B Results by dataset

Table 5 partitions the one global allocation by dataset. Each dataset cell contains 6,000 traces. Actions, Recovery, Damage, and Net are counts; EM and F1 differences are percentage-point differences from that cell’s Keep baseline, using the 6,000-trace denominator. No cell reallocates 900 replacements within its dataset. These descriptive partitions support no independent subgroup-superiority inference.

Table 5 Dataset breakdown. Each group has 6,000 traces; actions are partitions of one global 900-action allocation, not separately reassigned quotas. Differences from the group-specific Keep baseline are percentage points.

Policy	Actions	Recovery	Damage	Net	Δ EM (pp)	Δ F1 (pp)
2WikiMultiHopQA						
Keep	0	0	0	+0	+0.0000	+0.0000
HGB	292	136	12	+124	+2.0667	+2.6822
GbV	319	131	11	+120	+2.0000	+2.6189
ROA-FULL	298	134	8	+126	+2.1000	+2.8632
ROA-NOGBV	300	127	15	+112	+1.8667	+2.5163
HGB+GbV _R	297	129	4	+125	+2.0833	+2.8465
HGB-only _R	290	133	9	+124	+2.0667	+2.6644
GbV-only _R	318	132	10	+122	+2.0333	+2.6489
V2	313	139	10	+129	+2.1500	+2.8482
HotpotQA						
Keep	0	0	0	+0	+0.0000	+0.0000
HGB	323	132	16	+116	+1.9333	+2.6733
GbV	303	98	10	+88	+1.4667	+2.1477
ROA-FULL	315	122	10	+112	+1.8667	+2.6628
ROA-NOGBV	316	121	17	+104	+1.7333	+2.4339
HGB+GbV _R	309	121	7	+114	+1.9000	+2.7132
HGB-only _R	286	119	12	+107	+1.7833	+2.4747
GbV-only _R	300	99	10	+89	+1.4833	+2.1814
V2	317	123	14	+109	+1.8167	+2.5511

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Table 5 (continued)

Policy	Actions	Recovery	Damage	Net	ΔEM (pp)	$\Delta F1$ (pp)
MuSiQue						
Keep	0	0	0	+0	+0.0000	+0.0000
HGB	285	98	4	+94	+1.5667	+1.9603
GbV	278	89	4	+85	+1.4167	+1.7499
ROA-FULL	287	115	4	+111	+1.8500	+2.2084
ROA-NOGBV	284	107	6	+101	+1.6833	+1.9843
HGB+GbV _R	294	102	0	+102	+1.7000	+2.1124
HGB-only _R	324	105	5	+100	+1.6667	+2.0959
GbV-only _R	282	92	4	+88	+1.4667	+1.8109
V2	270	99	4	+95	+1.5833	+1.9524

Appendix C Results by retriever

Table 6 partitions the same global allocation by retriever. Each retriever cell contains 6,000 traces. Counts and percentage-point differences have the same meaning as in Appendix B, now using the retriever-cell baseline. The allocation is global, not separately optimized within these groups; these cells do not establish retriever-specific superiority.

Table 6 Retriever breakdown. Each group has 6,000 traces; actions are partitions of one global 900-action allocation, not separately reassigned quotas. Differences from the group-specific Keep baseline are percentage points.

Policy	Actions	Recovery	Damage	Net	ΔEM (pp)	$\Delta F1$ (pp)
BM25						
Keep	0	0	0	+0	+0.0000	+0.0000
HGB	386	149	15	+134	+2.2333	+2.9645
GbV	318	111	7	+104	+1.7333	+2.3527
ROA-FULL	291	119	4	+115	+1.9167	+2.5157
ROA-NOGBV	296	117	8	+109	+1.8167	+2.2911
HGB+GbV _R	300	118	2	+116	+1.9333	+2.5663
HGB-only _R	212	100	2	+98	+1.6333	+2.0095
GbV-only _R	295	109	5	+104	+1.7333	+2.3324
V2	391	154	10	+144	+2.4000	+3.0984
Dense						
Keep	0	0	0	+0	+0.0000	+0.0000
HGB	282	130	8	+122	+2.0333	+2.5925
GbV	276	101	6	+95	+1.5833	+2.0890
ROA-FULL	305	129	5	+124	+2.0667	+2.7886
ROA-NOGBV	307	122	15	+107	+1.7833	+2.4558
HGB+GbV _R	312	125	3	+122	+2.0333	+2.7335
HGB-only _R	395	150	13	+137	+2.2833	+3.0715
GbV-only _R	305	112	7	+105	+1.7500	+2.3119
V2	271	113	9	+104	+1.7333	+2.3713
Hybrid RRF						
Keep	0	0	0	+0	+0.0000	+0.0000
HGB	232	87	9	+78	+1.3000	+1.7588
GbV	306	106	12	+94	+1.5667	+2.0747
ROA-FULL	304	123	13	+110	+1.8333	+2.4302

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Table 6 (continued)

Policy	Actions	Recovery	Damage	Net	ΔEM (pp)	ΔF1 (pp)
ROA-NOGBV	297	116	15	+101	+1.6833	+2.1875
HGB+GbV _R	288	109	6	+103	+1.7167	+2.3724
HGB-only _R	293	107	11	+96	+1.6000	+2.1540
GbV-only _R	300	102	12	+90	+1.5000	+1.9970
V2	238	94	9	+85	+1.4167	+1.8821

Appendix D Policy construction details

D.1 Input features and fitting samples

The frozen numeric source order is: state-symmetric HGB; state-symmetric logistic; no-cross-state; no-B; no-evidence-change; no-answer-form; ordinary compact logistic; B-rule; higher-own-likelihood; likelihood margin; and paired GbV margin. The first seven are learned historical dependencies and the next three are fixed rules. The last source is defined in Eq. (1). Table 7 maps those sources to the five current Recovery heads.

Table 7 Current head construction. A flag accompanies each numeric score; all heads also contain three retriever indicators.

Policy	Numeric scores	Flags	Indicators	Width
ROA-FULL	Sources 1–11	11	3	25
ROA-NOGBV	Sources 1–10	10	3	23
HGB+GbV _R	HGB and paired GbV	2	3	7
HGB-only _R	HGB	1	3	5
GbV-only _R	Paired GbV	1	3	5

The documented fitter `src/arbitration/empirical_panel.py` forms sorted eligible-key sets separately within the fitting and calibration partitions before constructing numeric matrices. Eligible fitting rows determine medians, imputed means, and population standard deviations; a zero standard deviation becomes one. The nonfinite/missing flag is retained as a predictor, and the three retriever indicators are unscaled. The base head fits eligible fitting rows, and the Platt head fits eligible calibration logits.

The 3,600/900 question-group split defines the fitting/calibration populations before eligibility filtering; it does not give the numerical fit sizes. The eligible fitting/calibration totals are 2,572/630 rather than the partition totals 10,800/2,700. Their class counts are 2,015 negative and 557 positive for fitting, and 498 negative and 132 positive for calibration. All five current heads use these same rows and Recovery labels.

The fitted heads exclude no extra cases on the basis of fresh evaluation results. Their target is Recovery; same-correctness transitions are negative whenever otherwise eligible. Raw HGB, raw GbV, and V2 remain contextual policies. Beyond HGB and paired GbV, the legacy inputs are treated as frozen score dependencies. Their complete feature subsets and training specifications are not included in the available source release. The ROA-FULL comparison is consequently conditional on that saved score matrix, not evidence that any named historical component is necessary.

D.2 NLI input and chunking interface

The current documented scorer is `src/verification/gbv_nli.py`. Its `format_hypothesis` function interpolates the question and branch answer into the literal template printed in Sect. 3.2. Evidence passages are premises. Token lengths are checked on complete premise–hypothesis pairs with special tokens included. A premise already fitting the context is used as one chunk. For an overlength premise, word-window endpoints are searched to obtain fitting chunks; the next start retains up to 20 words of overlap while forcing forward

progress. The full hypothesis is unchanged. The scoring tokenizer uses `truncation=False` and pads only for batching.

The scorer applies softmax to FP32 logits, uses the exact positive entailment label rather than substring matching against “not_entailment,” and takes the maximum over all scored passage chunks for a branch. The project-pinned configuration resolves the positive index to zero and the context limit to 512 tokens. The numeric margin is the repaired-branch maximum minus the original-branch maximum. The file version defining this interface is identified in Sect. D.3.

D.3 Code versions and training metadata

The public source release is ReliableRAG-Code at Git commit `038d769e96d092a2eec3bbc5df9f45f3c2a17ff0`; the repository URL is <https://github.com/qianxingji/ReliableRAG-Code>. The following identifiers bind the documented implementation and numeric records to that release.

Current-head fitting and target. The fitting source is `src/arbitration/empirical_panel.py`, Git blob `f27090ee94a04c4fc693dd9d0a61380660db13e9`. The contract and target source is `src/arbitration/empirical_contract.py`, Git blob `60bd9a96ef27072e856f30a1f163b3c768f240d1`. The paired NLI source is `src/verification/gbv_nli.py`, Git blob `8be2e0b54e6ff0fce61aba361b4fa4785904b724`.

Development records and fitted parameters. The text-free bundle `DEVELOPMENT_NUMERIC.jsonl.gz` has SHA-256

8867daa8c8d827d5aa928627d8de068656e35760b47f784c38d328a369afe7f0

It contains 13,500 opaque-identity rows, including 2,572 eligible fitting rows (557 positive) and 630 eligible calibration rows (132 positive). The parameter file `CURRENT_HEADS.json` has SHA-256

a519b80b08ece87c59d296a3b590a1cda31de16122c68a1e0efe5a2c27f1e5ea

and records all preprocessing values, base coefficients/intercepts, Platt slope/intercept, iterations, target counts, and hashes. The workflow `scripts/reproduce_current_heads.py`, Git blob `9b150b1f1183b7e927b80c871c8b53cfda37ce9b`, refits all ten logistic models and compares them exactly with the accepted records.

Evaluation records and statistical reconstruction. The evaluation bundle has SHA-256

a930286d62bddf9f4837cb1beb4b863f0f94c1bc697b2311977a9b0eb18f09f7

The workflow `scripts/reproduce_paper_statistics.py`, Git blob `19322ba6dcb3bf818e4bcda123041bb6ee2c2ace`, reconstructs the reported point estimates and bootstrap results from all 18,000 evaluation rows.

For each base head of width d , the fitted model has d coefficients plus one intercept; the Platt model adds a slope and intercept. Thus ROA-FULL, ROA-NOGBV, HGB+GbV_R, HGB-only_R, and GbV-only_R contain 28, 26, 10, 8, and 8 learned scalar parameters, respectively, excluding stored preprocessing statistics.

Appendix E HGB-based pairwise preference scoring

E.1 Training target and feature representation

The HGB component is a histogram-based gradient-boosting model for pairwise preference scoring, trained in an earlier development stage using `HistGradientBoostingClassifier`. Its parameters are held fixed throughout the present study. Its output is used either directly for ranking or as an input to the current Recovery heads; it is distinct from the HGB-only_R head fitted in the current experiment. Its historical development universe contains 7,200 labeled traces from 4,800 question groups and 1,539 available pair-feature records. Only 601 informative traces from 518 question groups enter the final fit. For normalized exact-match outcomes $c_i \in \{0, 1\}$,

$$y_H = \begin{cases} 1, & (c_0, c_1) = (0, 1), \\ 0, & (c_0, c_1) = (1, 0). \end{cases} \quad (9)$$

Pairs with $(c_0, c_1) = (0, 0)$ or $(1, 1)$ have no preference label and are excluded. HGB thus learns a preference on informative transitions, not the current full eligible-population Recovery target.

For $i, j \in \{0, 1\}$, the likelihood cell is

$$L_{ij} = \frac{1}{T_i} \sum_{t=1}^{T_i} \log p_R(a_{i,t} \mid q, E_j, a_{i,<t}), \quad (10)$$

where only the fixed answer tokens contribute, not prompt positions. L_{00} and L_{11} are own-evidence likelihoods; L_{01} and L_{10} use the other branch’s evidence. Define

$$\phi_i = (\ell_i^{\text{own}}, \ell_i^{\text{cross}}, \ell_i^{\text{own}} - \ell_i^{\text{cross}}, n_i, u_i^+, u_i^-, s_i^+, s_i^-), \quad (11)$$

with $(\ell_0^{\text{own}}, \ell_0^{\text{cross}}) = (L_{00}, L_{01})$ and $(\ell_1^{\text{own}}, \ell_1^{\text{cross}}) = (L_{11}, L_{10})$. Here n_i is the stored answer-length descriptor; u_i^+ and u_i^- are multiset token-F1 compatibility with the added and removed document text; and s_i^+ and s_i^- are SequenceMatcher ratios after text normalization. The latter historical field names include “semantic compatibility,” but their implementation is string-sequence similarity, not a neural semantic model. The lexical tokenizer used by these stored feature descriptors is not specified in the available historical source release. These descriptors are therefore distinguished from the reader-token average in Eq. (10).

Let $\delta = \phi_1 - \phi_0$. The five pair-context variables are

$$z = (o_E, d_S, n_E, t_A, s_A), \quad x = [\delta; \delta \otimes z] \in \mathbb{R}^{48}. \quad (12)$$

o_E is Jaccard overlap of document IDs in E_0 and E_1 ; d_S is added-document retrieval score minus removed-document retrieval score; n_E is one minus the largest token-set Jaccard similarity between the added document and a document in E_0 ; t_A is token F1 between a_0 and a_1 ; and s_A is the saved BGE answer-pair similarity. The construction contains eight branch differences and forty difference–context interactions. Dataset and retriever identities are absent from this historical input.

E.2 Model specification and score symmetry

Each historical example is duplicated as (x, y_H) and $(-x, 1 - y_H)$. The fixed classifier uses `max_iter=300`, `max_leaf_nodes=15`, learning rate 0.08, L2 regularization 1.0, and final-fit seed 20261834; no feature scaling or hyperparameter search is applied to this final fit. If $p_H(1 \mid x)$ is its class-one probability, the stored score is

$$h(x) = \frac{1}{2} \{p_H(1 \mid x) + 1 - p_H(1 \mid -x)\}. \quad (13)$$

It follows algebraically that $h(-x) = 1 - h(x)$, up to numerical precision. A larger $h(x)$ favors a_1 . Raw HGB ranks directly by $h(x)$; HGB-only_R and HGB+GbV_R use that frozen score as an input to their current Recovery heads and do not refit HGB.

The historical “state-symmetric” identifier refers here to sign augmentation and the complementary score under $x \mapsto -x$. It does not prove that exchanging (a_0, E_0) with (a_1, E_1) and recomputing the representation yields exactly $-x$. Added/removed-document roles and directional context variables can change under that exchange; in particular, the score difference and novelty descriptors need not stay fixed. The claim is therefore restricted to representation-level negation, with no stronger raw-branch exchange invariance asserted.

E.3 Implementation provenance

The historical HGB feature source is publicly released as `src/mars/state_symmetric.py` at commit `038d769e96d092a2eec3bbc5df9f45f3c2a17ff0`; its SHA-256 is

3724b5ac77722b70cabdf2589379d5584942f34ec81f3f5a04f88e0665f8f717

The historical model-binding file `src/mars/full_experiment.py` has SHA-256

65713509ef6c8d094bb44ad2c47b0238ceade607616d3e729b31c56096326d50

and remains outside the public release.

The original saved HGB artifact is 639,652 bytes with SHA-256

9245170f855435b5603b04013bd4fbab79a75d2761853a012ab9f3259600bf6e

Source-assembled replay reproduced all 1,539 HGB scores with zero numerical error. The serialized replay differed only in saved CPU-thread metadata (one rather than six); observed learned arrays and tree fields matched. Original per-estimator fit-time ID/matrix receipts and an independent original-fit witness remain unavailable, so this supports saved-computation equivalence, not independent authentication of the original training event.

Appendix F Answer metrics

Normalization lowercases text, removes ASCII punctuation and the English articles “a,” “an,” and “the,” and collapses whitespace. EM is one if a normalized prediction equals any allowed reference. Token F1 uses multiset token overlap and takes the maximum over allowed references. Unequal answers involving “yes,” “no,” or “noanswer” receive zero F1; an empty token sequence scores one only against another empty sequence. The method-blind metric implementation follows these conventions. These outcome-metric conventions are distinct from the incompletely specified historical feature tokenization described in Appendix E.

Appendix G Data sources and development-set separation

The evaluation source frames are HotpotQA validation, 2WikiMultiHopQA development, and MuSiQue training. The last label denotes the public benchmark split, not membership in the current selector’s fitting set. A prospectively fixed hash order selects 2,000 questions per dataset after excluding the recorded 18,600-question union. The cohort is deterministic and frame-scoped; it is neither a probability sample nor a semantic-decontamination certificate.

The seven historical upstream estimators have reconstructed membership of 601 fitting traces from 518 question groups within a historical universe of 7,200 traces and 4,800 questions. Those 4,800 groups and 3,000 earlier V2 questions belong to the exclusion union and have zero recorded dataset/ID overlap with the current five-head development cohort. Only the five current heads share the 4,500-question, 3,600-fitting/900-calibration recipe. The historical totals describe a different development universe; they do not specify its per-dataset composition or a fully crossed retriever design. Matching current heads does not equalize every signal’s total historical label budget.

Appendix H Provenance and excluded model extensions

The study record preserves saved estimator bytes, reconstructed historical membership, numerical replay, pre-reference scoring witnesses, action memberships, outcome mapping, aggregate analysis, and independent validators under hashes. A locally preserved pre-ROA-named package contains a byte-identical historical training manifest; its timestamp has no independent external certification. Seven original per-estimator fit-time ID/matrix receipts and an independent original-fit witness remain unavailable. These records do not exclude unknown unlogged activity.

The Qwen/BGE replay covers a prospectively selected 180 of 18,000 traces. Downstream saved neural witnesses are checked, but this is not a second complete neural-forward run. A Phi extension acquired artifacts and failed the fixed semantic validation gate: 6,573 of 9,120 BF16 query vectors exceeded tolerance 10^{-3} , with maximum deviation 0.0041683525. It was excluded before policy fitting and fresh effect analysis. Mistral stopped at protocol review before engineering or model execution. Neither is a failed reader-effect experiment, and neither supplies robustness evidence.

Appendix I Computational workload details

Across the complete project history, 185 scientific fit calls were recorded; ten belong to the five current base/calibration bundles. The historical total does not alter the matched training conditions of the current five-head comparison.

Tables 8 and 9 retain the reported workload counts and timing boundaries. These are shared-graph disclosures, not standalone deployment measurements. Hybrid retrieval uses both retrieval components, so component counts differ from trace counts.

Table 8 Recorded shared-graph operation counts and token totals.

Component	Recorded value
Original / repair retrievals	18,000 / 18,000
Original retrieval components	12,000 BM25; 12,000 dense
Original embedding forwards / document rows	15,422 / 54,716
Repair BGE query forwards	12,000; no document re-embedding
Qwen generation receipts	54,000
Qwen prompt tokens	51,901,556
Qwen generated tokens / output-score steps	826,362
Base answer-embedding forwards	2,250
Base likelihood forwards / cache hits	16,932 / 136
Paired GbV NLI forwards / premise–hypothesis pairs	8,534 / 42,946

Table 9 Preserved stage timers. Values are seconds; the boundaries differ and are not additive.

Stage	Seconds
C2	1,096.562
C3	45,216.887
Base scoring	5,337.573
GbV scoring	3,523.022
Allocation	157.658

Separate saved C4 allocator peaks are 8,153,970,176 bytes for base scoring and 2,956,706,304 bytes for GbV scoring. They occur in different stages and must not be added. The bounded replay separately records 540 generation receipts, 518,966 prompt tokens, and 8,441 generated tokens/forwards. It is a reproduction witness, not an independent workload estimate.

Standalone policy and per-generation latency, a uniform end-to-end timer, canonical C2/C3 allocator peaks, simultaneous whole-system peak memory, C2/C3 BGE token totals, generation input-token touches per autoregressive forward, floating-point operation counts, and energy were not measured. Missing values are not zero; prompt/output-token totals are not estimates of floating-point operations. The global 5% allocation is not a 5% compute, latency, or memory budget.

Appendix J Scope of available replication materials

The public repository at <https://github.com/qianxingji/ReliableRAG-Code>, fixed for this manuscript at commit `038d769e96d092a2eec3bbc5df9f45f3c2a17ff0`, releases the current Recovery-head fitting procedure, accepted current-head parameter records, paired NLI scorer, historical HGB feature source, eligibility/allocation code, and two text-free numeric bundles.

Current-head reproduction. The 13,500-row development bundle retains opaque sibling groups, frozen fitting/calibration roles, eligibility, 11 numeric inputs, and binary EM outcomes. Running `scripts/reproduce_current_heads.py` refits five base logistic heads and five disjoint Platt calibrators and checks exact agreement with the accepted preprocessing, learned parameters, iterations, target counts, ranking metadata, and design hashes. The effective fitting/calibration sizes are 2,572/630, with 557/132 Recovery positives.

Statistical reproduction. The 18,000-row evaluation bundle retains opaque sibling groups, stable tie order, eligibility, scores, actions, and numeric EM/F1 outcomes. The complete public workflow rebuilds all nine-policy point estimates and the fixed-seed, 20,000-draw stratified question-cluster bootstrap, including repeated global top- K allocation and fixed-action sensitivity, with exact comparison to the saved results.

Upstream reconstruction boundary. The release excludes benchmark text and original IDs, references, generated answers, retrieved passages, model weights, historical fitted estimators, and private execution receipts. It cannot regenerate retrieval, candidate answers, neural scores, or historical upstream training. Incomplete historical generation and legacy-component specifications remain. The comparisons consequently condition on realized candidate pairs and saved scores; current-head and statistical reproduction do not remove this upstream limitation.